PondGate Detention-Pond Routing in Practice — User Manual and Worked Examples

PondGate v1.3.4 | English edition

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Every number in this book was produced by this software version and checked with an independent program.

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Preface

This book is the handbook of PondGate. It has two parts.

Part I is the user manual. It follows the order of the software screens and explains what to enter, what to press and what to read in each module. First-time users can start with Chapter 4, "Five-minute start", and look up the other chapters when needed.

Part II is a set of worked examples. It starts from the hydraulic principles of detention ponds. It then uses the public benchmark cases built into the software to work step by step through routing, interpretation, design screening and cross-checks with external models. Each chapter has four parts: principle, step-by-step example, check and exercises. Answers to the exercises are in Appendix C.

Scope of this book

PondGate computes the water level, storage and outflow of one detention pond under a known inflow hydrograph. It does not compute rainfall–runoff, channel routing, long-term continuous simulation or ponds in series.

Every result in this book holds only under the declared inputs and assumptions. Neither the software nor this book replaces design codes, site investigation or the judgement of a professional engineer. Every tick and record marked "teaching assumption" is for practice only and must not be used in real projects.

Where the numbers come from

Every result number in Part II was read from this version of the software after a real run. An independent small program then recomputed it for checking. If you get a different number with the same software version, first check that your inputs are exactly the same as in the book.

Version and citation

This English edition matches PondGate v1.3.4. The screenshots were taken in the English interface. The source code is public on GitHub (hongyaoming-

cloud/detention-pond-dss), and the Zenodo DOIs of each version are listed in Appendix D.

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Chapter 1 What PondGate can and cannot do

PondGate is detention-pond routing software that runs in a web browser. For one detention pond under a known inflow hydrograph, it computes how the water level, storage and outflow change over time. It keeps the inputs, method and results of every run as a traceable record.

1.1 The method in one sentence

The software solves the level-pool (storage continuity) equation:

> $dS/dt = Qin(t) - Qout(H)$

Here S is the storage in the pond, H is the water level, Qin is the inflow and Qout is the outlet discharge. The equation is solved with the fourth-order Runge–Kutta method at a fixed step (RK4, Δt = 2 s). Chapters 11 to 13 explain the principle and show hand checks.

1.2 What it can do

- Inflow–outflow routing of a single detention pond, giving the peak reduction, the maximum water level and the event-added detention volume.
- "Constrained design screening" and "continuous optimization" under a declared allowable outflow and allowable level, to find feasible pond areas and outlet sizes.
- Download of a traceable report with the inputs, results and scope of application.
- Numerical comparison with HEC-HMS and SWMM results.
- After signing in, saving a case as a project and sharing it with colleagues in the same company (Chapter 2).

1.3 What it cannot do

- Rainfall–runoff calculation: the user must supply the inflow hydrograph.
- Channel routing, long-term continuous simulation or ponds in series.
- Field calibration: agreement with other models is called "numerical consistency" only.

- Engineering approval: results are not a final design. Engineering responsibility stays with qualified professionals.

1.4 The software will not say "pass" for you

When no feasible design exists within the current limits and search bounds, PondGate shows "Blocked". It lists the dominant limits and the directions for revision, and it does not show "pass". This is deliberate: it stops users from getting a result that looks acceptable when the evidence is not enough. Chapter 18 uses an example to show how to read a blocked result.

Chapter 2 Accounts, companies and projects

You must sign in to use the official PondGate website. Your company administrator creates your account. After signing in, you have all calculation functions, and you can also save a case as a project, continue it later and share it with colleagues in the same company. A teaching or demonstration version without sign-in has the same calculation functions but no project storage.

2.1 Four roles

Role	Can do	Cannot do
System administrator	Add companies and accounts for each company	Open company projects
Company administrator	Manage accounts of their own company and view all company projects	Manage other companies
Engineer	Create and save their own projects and share them with colleagues	Change colleagues' projects
Viewer	Open projects that colleagues have shared	Save

The system administrator cannot open company projects on purpose. This keeps the engineering data of each company inside that company.

2.2 First sign-in

1. Press "Sign in" at the top right of the screen.
2. Enter the user name and initial password that your administrator gave you.
3. The system asks you to change to your own password at once. The new password needs at least 8 characters and must be entered twice.
4. If you forget your password, ask your company administrator to reset it. The system does not send passwords.

2.3 Order for creating accounts

1. The system administrator first adds the company.
2. The system administrator creates one "company administrator" account for that company.

3. The company administrator signs in and creates the engineer and viewer accounts.

When creating an account, you can press "Random password" and then give the password to your colleague in private.

2.4 Save and open projects

- Engineers and company administrators can press "Save / open projects" to save all current inputs as a new project, with a name and a note.
- If you tick "Share with my organisation (read-only for others)", colleagues can open the project but can only view it, not change it.
- The current project name is shown at the top of the screen. When "● unsaved changes" appears, remember to press "Save".
- After you open a project, results do not appear automatically. Go to Module 4 and press "Run routing and evaluation" to compute again. This makes sure that the results always come from the current software version.

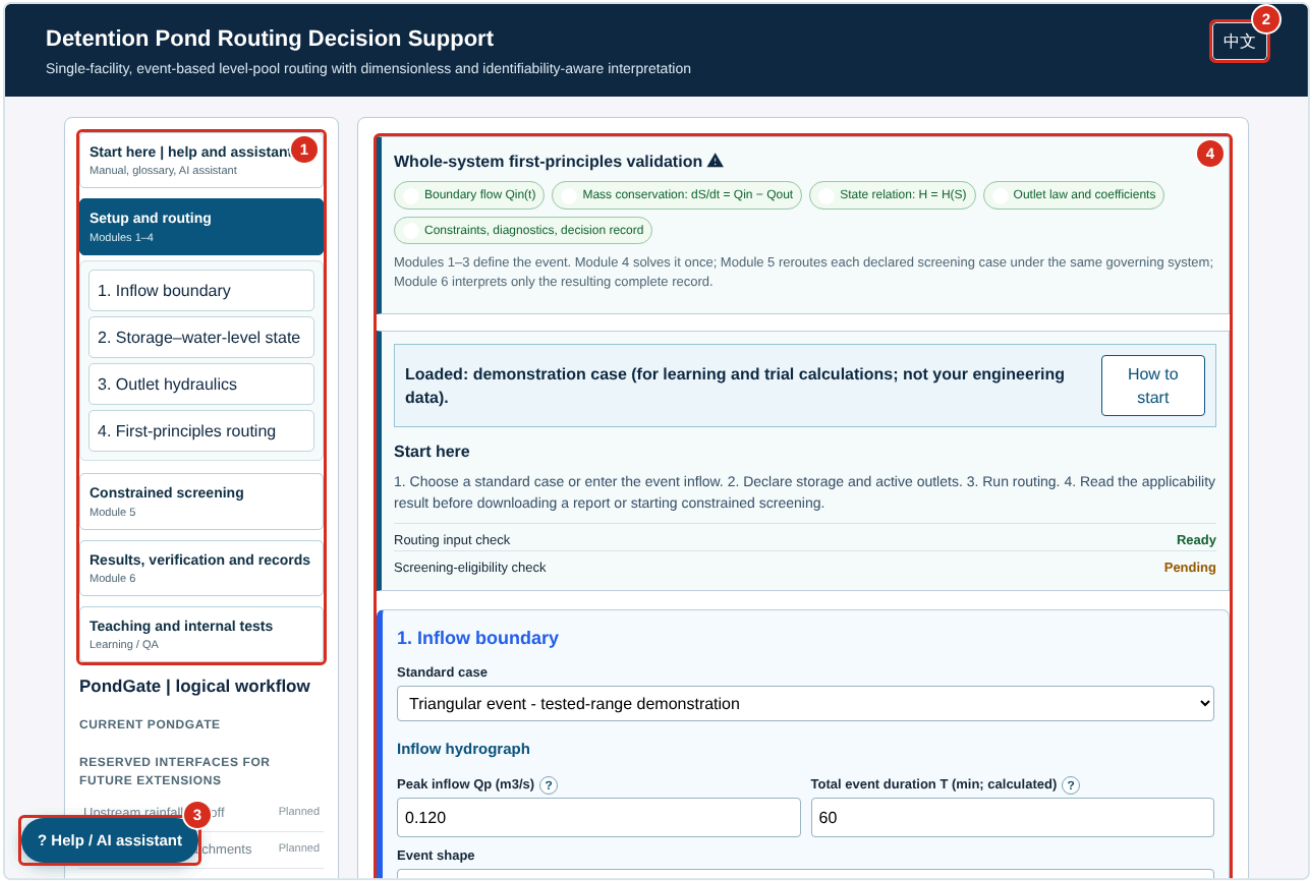
2.5 Recycle bin and activity log

- A deleted project only moves to the recycle bin; it is not really removed. Press "Restore" in the recycle bin to get it back.
- A company administrator can see the latest 300 actions under "Accounts", for example who created, shared or deleted which project and when.

2.6 Where the data are stored

All calculations run inside your browser, and no data are sent during a calculation. Only when you press "Save" are the input values of the project stored in the PondGate cloud database (Cloudflare). Passwords are kept only as encrypted hashes, and nobody (not even the system administrator) can see your password.

Chapter 3 Screen tour

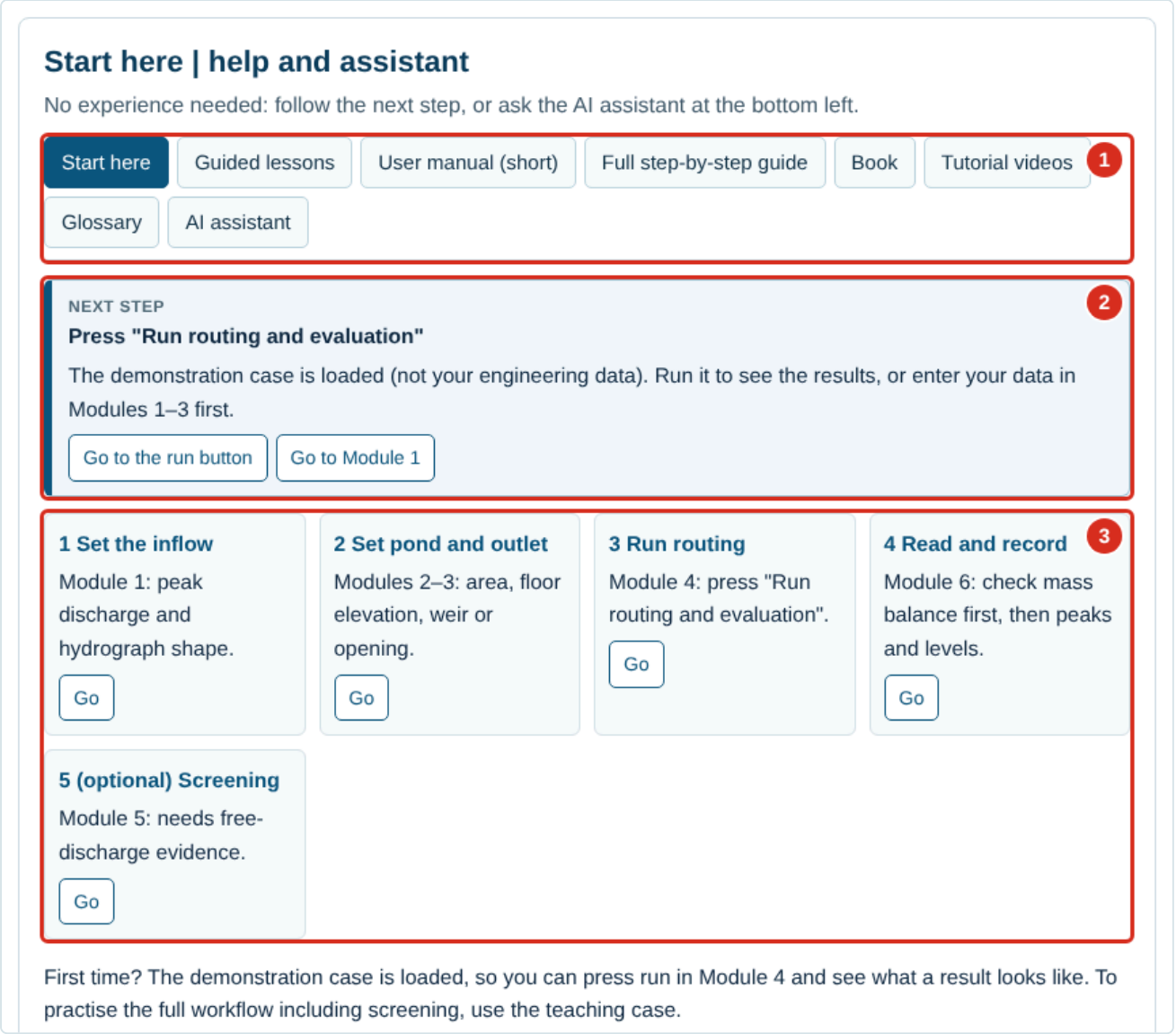


Screen layout of PondGate

Figure 3-1 shows the four main areas of PondGate.

1. Tab bar: "Start here", "Setup and routing", "Constrained screening", "Results, verification and records" and "Teaching and internal tests".
2. Language button: when it shows "中文", the interface is in English.
3. "? Help / AI assistant" button, available at any time.
4. Work area: the content of the current page.

3.1 The "Start here" help centre



The "Start here" page

Figure 3-2 shows the "Start here" page. The tabs at the top are, in order: Start here, Guided lessons, User manual, Full step-by-step guide, Book (this book), Tutorial videos, Glossary and AI assistant.

- The "Next step" box changes as you work and tells you what to do now.
- "Guided lessons" has 7 lessons from easy to harder. The system ticks each step when you finish it.
- The "?" next to a field name explains the term.

3.2 Tips

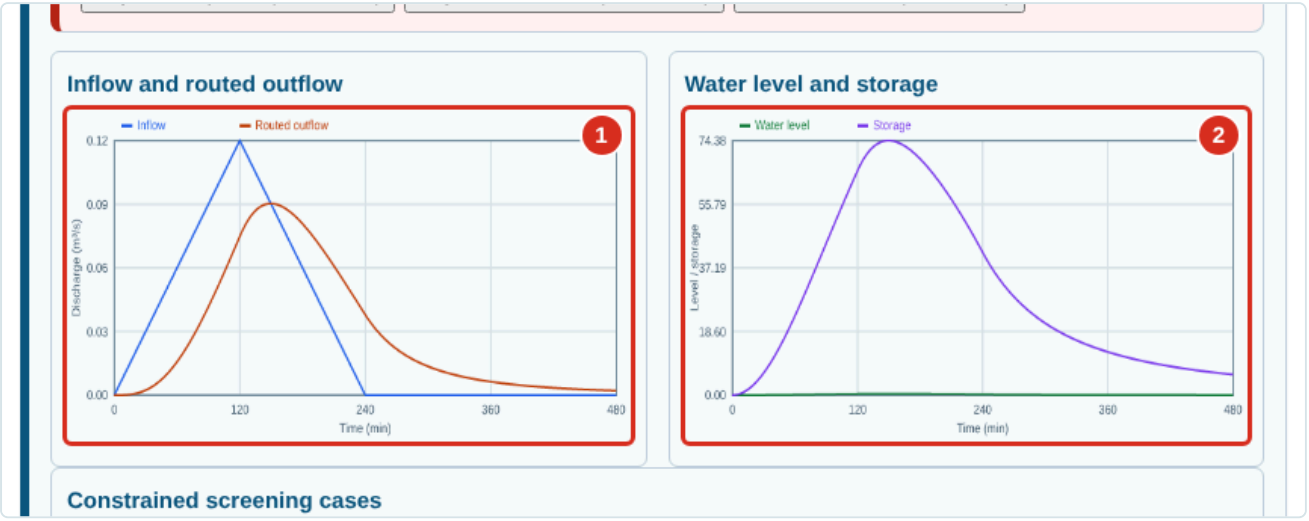
- Each time you press a button, a message at the bottom right explains what just happened.

- Switching the language does not change the input values. If the inputs changed after the last run, switching the language reruns the calculation with the current inputs, and the screen warns you first.
- The software also works on a phone. The tab bar then moves to the top of the screen.

Chapter 4 Five-minute start

This chapter uses the "triangular event" demonstration case, which is loaded when the software opens, to complete your first routing run. The demonstration case is for practice only and does not stand for any real project.

1. Open PondGate and switch to English if needed. Check that the tab bar is on "Setup and routing".
2. Scroll down to Module 4 at the bottom of the page and press "Run routing and evaluation".
3. The page switches to "Results, verification and records".
4. In the "Beginner check" box above the results, read what the system says about Hmax and Qp,o.
5. In the results table, first check that the "Mass-balance residual ε/Vin" is close to 0 %. Then read the peak inflow, the peak outflow and the peak water level.
6. Press "Download reproducible decision-support report" to save the record of this run to your computer.



Inflow and routed outflow hydrographs

Figure 4-1 shows the inflow and outflow hydrographs of the demonstration case. The outflow peak is lower than the inflow peak and comes later. This is the "peak reduction" and "delay" effect of a detention pond.

Table 4-1 lists the numbers you should see. If the numbers are the same, you did it right.

Item	Value you should see
Peak inflow Qp,i	0.1200 m³/s
Peak outflow Qp,o	0.0903 m³/s
Peak water level Hmax	0.413 m
Mass-balance residual ε/Vin	0.000 %
Beginner check	Not good enough

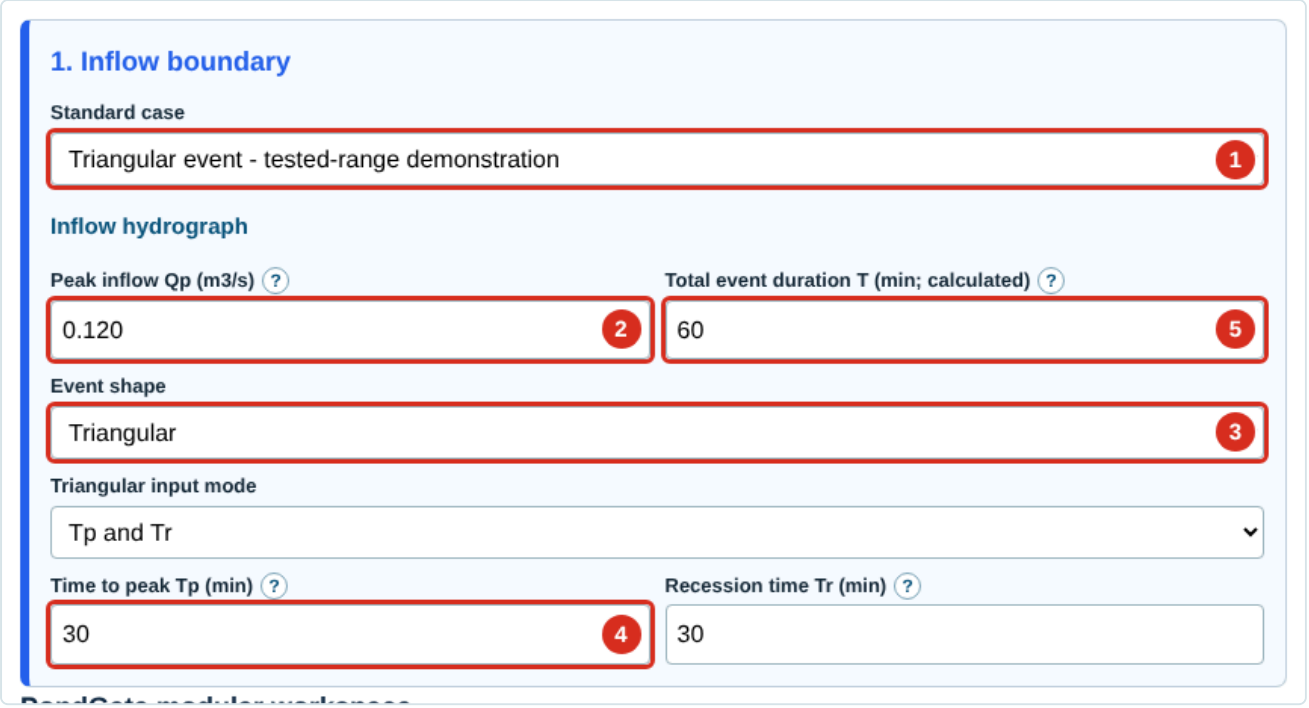
The default allowable outflow Qallow of the demonstration case is 0.080 m³/s. The peak outflow of 0.0903 m³/s is above this value, so the "Beginner check" says "Not good enough". This is the expected result, not an operating error. Chapters 10 and 14 explain the reason and how to improve the design.

There are two ways to reset all inputs to the demonstration case. First, open "Start here → Book" and press "Back to the base case". Second, reload the web page. Both clear any unsaved changes. After a reload, switch to English again if the page opens in Chinese.

Readers who want to follow every step with a screenshot can open "Start here → Full step-by-step guide", Chapter 3.

Chapter 5 Modules 1 to 3: describe the inflow, the pond and the outlet

5.1 Module 1: inflow boundary



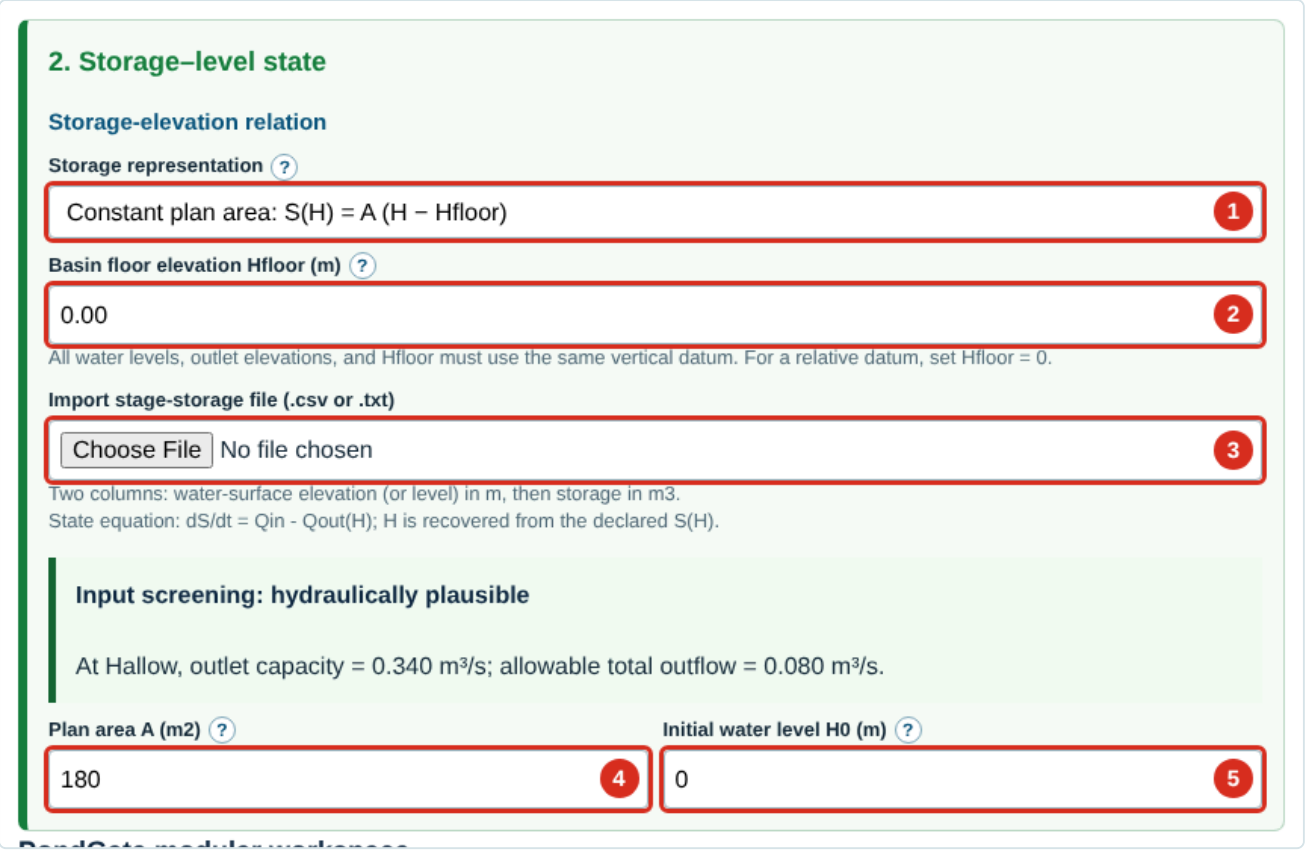
Module 1 inflow boundary

Figure 5-1 shows Module 1. The inflow can be described in three ways.

- Triangular event: enter the peak inflow Qp, the time to peak Tp and the recession time Tr.
- Long-duration event: enter the rise, plateau and recession times to form a trapezoidal hydrograph.
- Imported time–discharge pairs: one "time (min), discharge (m^3/s)" pair per line. The first time must be 0, times must increase, discharges must not be negative, and at least two lines are needed.

The total event duration T is the sum of the parts and cannot be entered directly. Remember that the peak discharge alone is not enough: the duration and shape of the hydrograph also change the detention volume needed (Chapter 16 shows an example).

5.2 Module 2: stage–storage relation



Module 2 storage–water-level state

Figure 5-2 shows Module 2. Storage can be described in two ways.

- Constant plan area: S(H) = A × (H − Hfloor). Only the area A is needed.
- Tabulated stage–storage relation: import a two-column .csv or .txt file, with the water level (m) in the first column and the storage (m^3) in the second. Routing does not extrapolate beyond the table.

The floor elevation Hfloor, the initial level H0 and all outlet elevations must use the same datum. With a relative datum, set Hfloor = 0.

5.3 Module 3: outlet hydraulics and free-discharge evidence

3. Outlet hydraulics

Outlet relation

- [ ] Vertical riser shaft

Outlet hydraulic boundary

- [ ] Free-discharge boundary confirmed (?) 4

Required evidence for constrained screening

- [ ] Tailwater remains below every controlling outlet.
- [ ] No downstream backwater or pipe-friction control modifies Qout(H).
- [ ] No outlet is submerged over the routed water-level range.
- [ ] No gate, pump, or operational control changes the stated outlet law.

These four checks document why Qout = Qout(H) is applicable. Until all are checked, routing remains exploratory and constrained screening is locked.

Free-discharge evidence record (required for constrained screening) (?)

Drawing / field inspection / calculation record identifie 5

Screening-limit basis (required for constrained screening)

Source and applicability of Qallow, Hallow, Td, and

These records are preserved in the run report. They document user-provided evidence; the software does not independently verify tailwater, backwater, or project design criteria.

Outlet discharge-coefficient basis

Coefficient mode

Fixed coefficient

Weir Cw preset

Taiwan practice—sharp-crested rectangular weir (Cw = 1.70)

Orifice Cd preset

Standard sharp-edged orifice (Cd = 0.62)

Riser inlet Cr preset

Standard sharp-crested inlet (Cr = 1.70)

Riser shaft Cd preset

Standard sharp-edged shaft (Cd = 0.62)

Fixed values are preliminary screening settings. Select field-calibrated value to enter a project-specific coefficient.

▸ **Coefficient references and applicability**

- [ ] Circular pipe outlet
- [x] Sharp-crested weir 1

Weir coefficient Cw (?)

1.70 2

Weir width b (m) (?)

0.20

Weir crest elevation (m) (?)

0

- [ ] Rectangular opening 3

Opening coefficient Cd (?)

0.62

Opening area Ao (m2) (?)

0.010

Opening height a (m)

0.10

Opening width b = Ao/a (m)

0.100

Opening invert elevation (m) (?)

0.05

Figure 5-3 shows Module 3. Only two outlet types can enter formal constrained screening.

- Sharp-crested weir: Qout = Cw × b × (H − P)^1.5. Enter the weir coefficient Cw, the weir width b and the crest elevation P.
- Rectangular opening: enter the discharge coefficient Cd, the opening area Ao, the opening height and the invert elevation. Before the water rises above the top of the opening, the opening acts like a narrow weir. After that, the discharge increases with head according to the orifice equation. Chapter 12 gives the full equations.

Risers, circular pipes, stage-dependent coefficient tables and combined outlets (weir and opening together) are for "research inspection". You can route them and read the results, but constrained design screening is locked.

You can use the default coefficients (Cw = 1.70, Cd = 0.62) or enter calibrated site values. The defaults are suitable only for preliminary screening.

The free-discharge evidence consists of one main tick, four checks (tailwater, backwater and pipe friction, submergence, operational control) and two text records. Tick them only when you really have drawings, site checks or calculations to support them. The software does not verify these conditions for you.

Chapter 6 Modules 4 and 6: run the routing and read the results

6.1 Run

Press "Run routing and evaluation" in Module 4 to start the calculation. If the button is grey, a field in Modules 1 to 3 has an unreasonable value. The fields to fix are listed above the button. Click one to jump to it; it is outlined in red.

Common causes are: Qp less than or equal to 0, an area of 0, a weir width or coefficient of 0, an outlet elevation below the floor, or no active outlet.

6.2 Beginner check

The "Beginner check" box at the top of the results page compares three results directly with the limits you declared: the peak water level Hmax with the allowable level Hallow, the peak outflow Qp,o with the allowable outflow Qallow, and the residual after the drain-down time Td with the allowable residual percentage.

- Green: all three are below their limits.
- Yellow: a value equals its limit and there is no safety margin. Treat it with caution.
- Red: a limit is exceeded. The box suggests a direction: a larger pond area usually lowers both the peak water level and the peak outflow; a smaller outlet lowers the peak outflow but raises the peak water level.

The beginner check only compares readings. It is not an engineering approval.

6.3 Order for reading the results table

Output	Value	Engineering meaning
Peak inflow Qp,i	0.1200 m³/s	Declared hydraulic loading
Peak outflow Qp,o	0.0903 m³/s	Sum of active outlet components 1
Peak ratio RQ = Qp,o/Qp,i	0.753	Peak attenuation descriptor 2
Event-added detention volume ΔSmax	74.385 m³	Maximum storage above the initial storage S0. 3
Drain-down residual at Te + Td	0.03 %	Declared limit = 1.00% of ΔSmax.
Storage ratio RS = ΔSmax/Vin	0.344	Event-added storage relative to inflow volume
Normalized lag RT	0.124	(tp,o - tp,i) / event duration
Peak water level Hmax	0.413 m	Compare with allowable level and freeboard 4
Absolute conservation residual	1.259e-10 m³	Absolute difference between cumulative net inflow and storage change
Mass-balance residual ε/Vin	0.000 %	Relative numerical conservation check 5
Near-peak flatness FPI95	0.006	95% near-peak duration / routed-record duration
Median-volume time	42.6 min	Bulk timing measure
Discharge-centroid time	49.3 min	Bulk timing measure

Results table

Figure 6-1 shows the results table. Read it in this order.

1. Mass-balance residual ε/Vin: it should be close to 0 %. Above 0.5 %, the system asks you to solve the numerical problem first, and the results must not be used.
2. Peak outflow Qp,o and peak ratio RQ = Qp,o ÷ Qp,i: a smaller RQ means more peak reduction.
3. Peak water level Hmax and event-added detention volume ΔSmax: compare them with the allowable level and the freeboard.
4. Near-peak flatness FPI95: a large value means a very flat peak. A single peak time is then not reliable; use the volume-based timing measures instead.

6.4 When the inputs change, the results expire

If an input changes after a run, the screen marks the results as out of date. Press "Run routing and evaluation" again before you interpret them. Ticking the free-discharge evidence or filling in the record fields does not make the routing results expire, because these fields do not change the computed numbers.

Chapter 7 Module 5: constrained design screening and continuous optimization

Constrained design screening answers this question: within the limits and search range I give, is there a pond area and outlet size that meets all limits at the same time?

7.1 When screening is available

Constrained screening is available only when all of the following hold.

- Only one sharp-crested weir or one rectangular opening is active.
- The coefficients are fixed values.
- The main free-discharge tick, the four checks and the two records are all complete.

If a condition is missing, Module 5 lists what is missing, where to check it and what to enter. These conditions are usually confirmed by the responsible design engineer from drawings or site data.

7.2 Limits to enter

5. Constrained design screening

Design screening and constrained routing

RQ = 0.753; RS = ΔSmax/Vin = 0.344; RT = 0.124; ΠH = 0.500; ΠQ(Href) = 2.833; ΠS,Δ(Href) = 0.833; n = 1.5; Href = 1.000 m; Tp/Tr = 1.000. ΠS,Δ = [S(Href) − S0]/Vin is the declared event-added storage control. These are not universal design equations.

Allowable peak outflow (m3/s) (?)

0.080 (1)

Allowable water level (m) (?)

1.00

Outlet configuration for constrained screening

Formal screening is limited to one passive weir or one rectangular opening under confirmed free discharge. Other choices are retained only for research inspection and keep screening locked.

Passive weir only — formal screening (2)

Screening resolution

Standard discrete screen: up to two outlet dimensions

Screening objective

Minimum event-added detention volume ΔSmax

Dimensionless reference water level Href (m) (?)

1.00

Href must use the same datum as H, lie within the declared S(H) domain, and be no lower than H0. It defines ΠQ and ΠS; it is not automatically a safety limit.

Maximum drain-down time Td (h) (?)

24

Permitted residual event storage (%)

1

At Te + Td, event-added storage must not exceed the stated residual percentage of ΔSmax. Every screening case uses the same state equation, constraints and time step.

☑ Screen weir width b

Minimum b (m)

0.05 (3)

Maximum b (m)

1.00

☐ Screen opening area Ao

Minimum Ao (m2)

0.005

Maximum Ao (m2)

0.050

☐ Screen riser diameter D (research inspection only)

Minimum D (m)

0.15

Maximum D (m)

0.60

Standard search: up to two outlet dimensions; other active dimensions remain fixed.
Current Outlet-module dimensions (fixed unless selected for search): b = 0.20 m; Ao = 0.010 m²; D = 0.30 m.

Advanced continuous optimization | global candidate search + local refinement

The formal mode can search two continuous variables: plan area A and the size of one passive outlet (weir width b or opening area Ao). Every candidate reruns the same authoritative RK4 and never writes back to the working inputs. The result is not a proven continuous global optimum and is not final engineering-design validation.

Continuous minimum plan area A (m²)

60

Continuous maximum plan area A (m²)

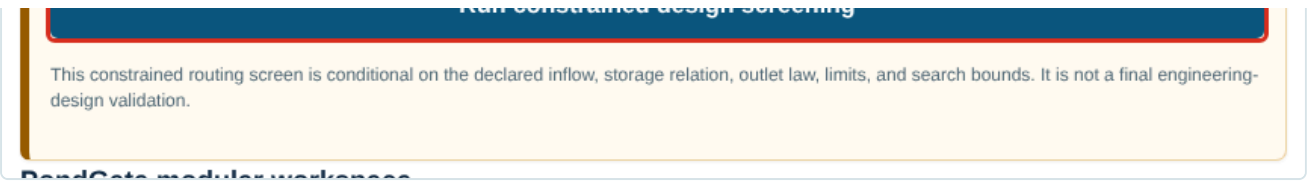
600

Continuous mode ready: A + b will be searched with the authoritative RK4 router.

▸ **Declared screening bounds**

(4)

Run constrained design screening



Module 5 constrained-screening settings

Figure 7-1 shows the settings of Module 5.

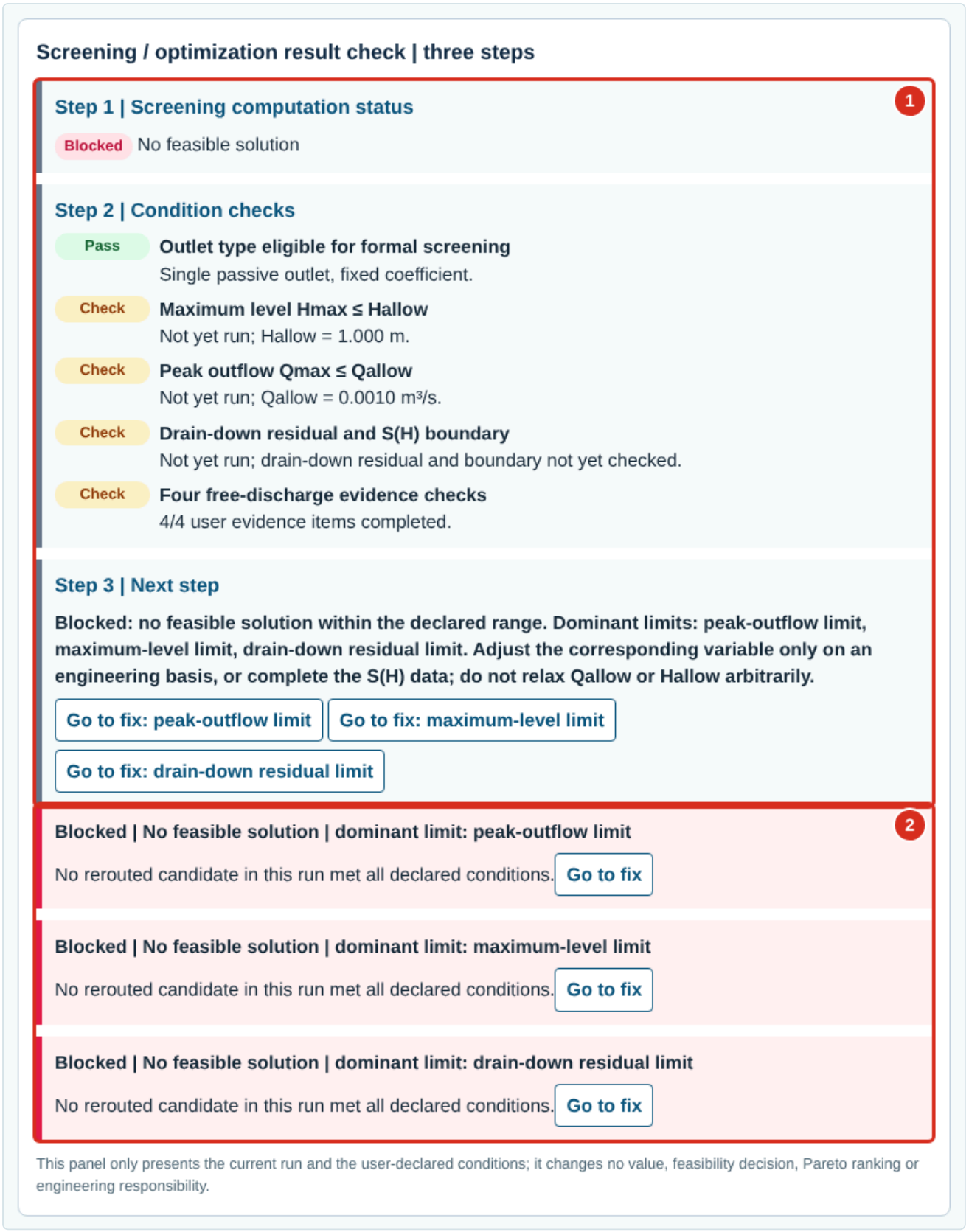
- Allowable peak outflow Qallow (m^3/s) and allowable water level Hallow (m): these should come from regulations or design criteria and be written in the "Screening-limit basis".
- Maximum drain-down time Td and permitted residual percentage: at the event end time Te plus Td, the event-added storage must not exceed this percentage of ΔSmax.
- Reference level Href: used only to compute the dimensionless parameters ΠQ and ΠS. It is not a safety limit.
- Search bounds: the minimum and maximum of the weir width b or the opening area Ao. For each ticked size, five equally spaced values are tested within the bounds.

7.3 Run and results

Press "Run constrained design screening". The page then switches to the results page. When a feasible case is found, the result lists the selected pond area, the outlet size and a comparison with each limit.

Continuous optimization searches the area A, between its minimum and maximum, together with one outlet size, and then refines locally. Its result is not a proven global optimum.

7.4 No feasible case



Blocked diagnosis when no feasible case exists

Figure 7-2 shows the screen when no feasible case exists. The system shows "Blocked" and the dominant limits (peak outflow, maximum water level or drain-down residual), with "Go to fix" buttons.

Change the pond area or outlet size on an engineering basis, or widen the search bounds where there is a basis for it. Do not relax Qallow or Hallow just to get a solution.

Chapter 8 Reports and traceable records

- The "Run record for this session (read-only)" on the results page lists the inputs, method, results and limits of this run.
- "Download reproducible decision-support report" saves a text file with the inputs, results and scope of application. You can attach it to the design record.
- When the inputs change, the old results expire. Run again before downloading.
- On the "Start here → Book" tab you can download this book as a PDF or a single web page, and the input data of the Part II examples.

Chapter 9 Error messages, common questions and the AI assistant

9.1 Common questions

Situation	Cause	What to do
The run button is grey	A field in Modules 1 to 3 is unreasonable	Read the red list above the button and click a field name to fix it
Constrained screening stays locked	More than one outlet, coefficients not fixed, or incomplete free-discharge evidence	Read the list of missing conditions in Module 5
"No feasible case" appears	No design within the bounds meets all limits	Change the area or outlet according to the dominant limit; see Chapter 18
Mass-balance residual above 0.5 %	A numerical problem, often with extreme inputs	Check the input units and the datum; see Chapter 13
Results are marked out of date	Inputs changed after the run	Press "Run routing and evaluation" again

9.2 AI assistant

Press "? Help / AI assistant" at the bottom left to open the AI assistant. It is an offline help assistant. It runs inside your browser, does not connect to any outside AI service and sends no data.

The AI assistant can do three things.

1. Answer operating questions, for example "What should I do next?" or "Why can't I press the button?".
2. Explain terms, for example "What is Qallow?".
3. Search this book: type a keyword (for example "orifice", "RK4" or "no feasible"), and it lists the most relevant sections. Click one to jump to it.

The AI assistant does not change your inputs and makes no engineering judgements. It can tell you how the numbers in the results table compare with the limits, but it will never say "this design can be adopted".

Part II Worked Examples

Each chapter of Part II follows the order "principle → step-by-step example → check → exercises". The inputs of every example are listed in tables, so you can enter them in the software and get the same numbers as the book.

Most of Part II starts from the demonstration case that is loaded when the software opens. Below it is called the "base case":

Item	Value
Inflow	Triangular, Qp = 0.120 m³/s, Tp = 30 min, Tr = 30 min
Pond	Constant plan area A = 180 m², floor Hfloor = 0, initial level H0 = 0
Outlet	Sharp-crested weir, Cw = 1.70, b = 0.20 m, crest P = 0
Drain-down condition	Td = 24 h, permitted residual 1 %
Design limits	Qallow = 0.080 m³/s, Hallow = 1.00 m

When the book says "start from the base case", first reset the inputs to the values in the table above. The easiest way is to open "Start here → Book", press "Back to the base case", and then press "Run routing and evaluation" in Module 4. Reloading the web page has the same effect (switch to English again if needed). Both ways clear unsaved changes, so save any case you want to keep as a project first.

Chapter 10 The role of a detention pond and the design limits

10.1 Principle

After development, the ground surface is less permeable, so the same storm produces a larger and faster flood peak. A detention pond stores part of the water and releases it slowly, so the peak seen downstream becomes smaller and later.

A design usually has two limits.

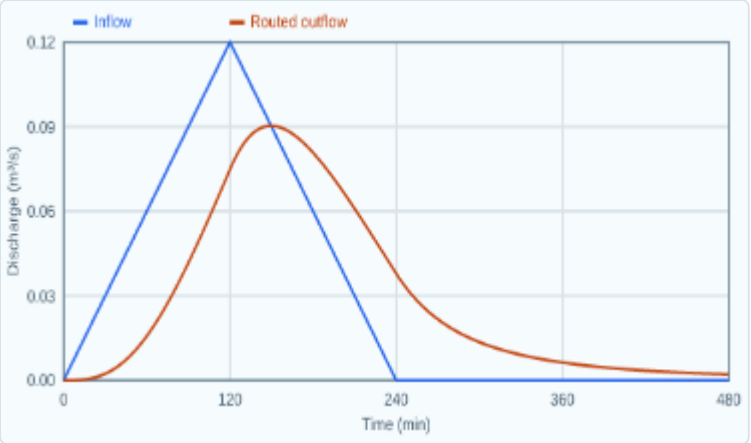
- Allowable outflow Qallow: the largest outflow allowed by the downstream drainage system or by regulations.
- Allowable water level Hallow: the upper limit of the water level in the pond, usually set by the pond depth, the freeboard or the surrounding ground levels.

A design is "good enough" only when Qp,o ≤ Qallow and Hmax ≤ Hallow at the same time, and the event-added storage left in the pond after the required drain-down time Td does not exceed the declared percentage.

10.2 Example: is the base case good enough?

This example assumes Qallow = 0.080 m^3/s and Hallow = 1.00 m (teaching assumption).

1. Open PondGate and keep the base case (if you have changed any input, press "Back to the base case" first).
2. In Module 5, enter Qallow = 0.080 and Hallow = 1.00.
3. Go back to Module 4 and press "Run routing and evaluation".



Inflow and outflow hydrographs of the base case

Figure 10-1 shows the inflow and outflow hydrographs of the base case. The results page reads as follows.

Item	Result	Limit	Reading
Peak outflow Qp,o	0.0903 m³/s	Qallow = 0.0800 m³/s	Exceeds
Peak water level Hmax	0.413 m	Hallow = 1.000 m	Within
Peak ratio RQ	0.753	—	About 25 % peak reduction
Drain-down residual (Te + Td)	0.03 %	1.00 %	Within
Event-added detention volume ΔSmax	74.385 m³	—	—

The "Beginner check" box says "Not good enough", because the peak outflow exceeds the allowable value. The water level still has a large margin. This means the pond is not too small; the outlet is too large and releases water too fast. Chapter 14 uses the same case to find a direction for improvement.

10.3 Key points

- To judge "good enough", read the outflow, the water level and the drain-down together. Reading only one of them can mislead you.
- When one limit is exceeded and another has a large margin, the pond area and the outlet are usually not well balanced.

10.4 Exercises

1. Change Qallow to 0.100 m³/s. Is the base case good enough?
2. Why does Qp,o exceed the limit although Hmax is only 0.413 m? Explain in one sentence.

Chapter 11 Hydraulic basics: the continuity equation and storage

11.1 Principle

The change of the water volume in the pond equals the inflow minus the outflow:

$$dS/dt = Qin(t) - Qout(H)$$

With a constant plan area, the storage is proportional to the water level:

$$S(H) = A \times (H - Hfloor)$$

Together these two equations are called "level-pool routing". It assumes that the water surface is level at every moment. It suits detention ponds with a large area, shallow water and low velocity.

11.2 Hand calculation: inflow volume

The volume of a triangular hydrograph is the area of the triangle:

$$Vin = ½ \times Qp \times (Tp + Tr) = ½ \times 0.120 \times 3600 = 216\ m^3$$

The software computes an event-added detention volume ΔSmax = 74.385 m³, so the storage ratio is RS = 74.385 ÷ 216 = 0.344, the same as in the results table. In other words, at most about one third of the inflow volume was in the pond at the same time.

11.3 An important property: the outflow peak lies on the falling limb of the inflow

When the outflow depends only on the water level in the pond (free discharge), dS/dt = 0 at the moment of the highest level, so Qin = Qout. Therefore the outflow peak always lies on the falling limb of the inflow hydrograph.

Check with the base case: an independent program puts the outflow peak at t = 37.4 min, with Qout = 0.0903 m³/s. The inflow at that moment is:

> $Q_{in} = 0.120 \times (60 - 37.4) \div 30 = 0.0904\ m^3/s$

The two differ by less than 0.0001 m³/s; the difference comes from rounding the time to 0.1 min. In Figure 10-1 the highest point of the outflow curve touches the falling line of the inflow. This is the property at work.

11.4 Key points

- The moment of the highest water level is the moment of the largest outflow, and also the moment when Qin = Qout.
- If another program shows an outflow peak that is not on the falling limb of the inflow, first check whether the outlet is affected by downstream conditions.

11.5 Exercises

1. If Tp = Tr = 45 min and Qp is unchanged, what is the inflow volume?
2. With A = 180 m², how much water-level rise does ΔSmax = 74.385 m³ represent? Compare it with Hmax.

Chapter 12 Outlet equations: weirs and rectangular openings

12.1 Sharp-crested weir

> Qout = Cw × b × (H − P)^1.5

In the base case, Cw × b = 1.70 × 0.20 = 0.34. Table 12-1 lists the discharge at different heads.

Head H − P (m)	0.10	0.20	0.30	0.40	0.50
Weir flow Qout (m³/s)	0.0108	0.0304	0.0559	0.0860	0.1202

The discharge grows with the 1.5 power of the head, so a small rise in level greatly increases the outflow.

12.2 Rectangular opening

PondGate computes a rectangular opening in two ranges. The opening height is a, the width is w = Ao ÷ a, and the head is y = H − invert elevation.

- The water has not risen above the top of the opening (y ≤ a): the opening acts as a weir of width w, Q = Cw × w × y^1.5.
- The water is above the top of the opening (y > a): the discharge starts from the weir flow at the top of the opening and adds the increase given by the orifice equation. The discharge is therefore continuous at the transition point and does not jump:

> Q = Cw × w × a^1.5 + Cd × w × a × [√(2g(y − a/2)) − √(2g × a/2)]

Take Ao = 0.010 m², a = 0.10 m, invert 0.05 m, Cw = 1.70 and Cd = 0.62 as an example. Table 12-2 compares the discharges of the weir and the opening.

Water level H (m)	0.15	0.30	0.50	0.75	1.00
Weir b = 0.20 m (m³/s)	0.0198	0.0559	0.1202	0.2208	0.3400
Opening Ao = 0.010 m² (m³/s)	0.0054	0.0115	0.0166	0.0214	0.0253

The discharge of the opening grows only with about the 0.5 power of the head. As the level rises, a weir releases water quickly, while an opening releases it slowly. This difference drives the design behaviour in Chapters 14 and 15.

12.3 Coefficients and free discharge

Cw = 1.70 and Cd = 0.62 are common preliminary values. The coefficients of a real structure depend on its shape, edges and approach velocity. A formal design should use calibrated values.

All these equations assume free discharge: the downstream water level is low enough not to back up into the outlet. If the outlet may be submerged from downstream, the equations no longer hold. This is why constrained screening requires the free-discharge evidence.

12.4 Exercises

1. If the weir width is doubled, by what factor does the weir flow change at the same head?
2. When the water level at the opening rises from 0.50 m to 1.00 m, by what percentage does the discharge increase? And for the weir?

Chapter 13 Numerical method: RK4, time step and mass balance

13.1 Principle

The continuity equation has no general analytical solution, so it is integrated numerically step by step. PondGate uses the fourth-order Runge–Kutta method (RK4), which computes four slopes in each step:

$$k1 = f(t, S)$$
$$k2 = f(t + \Delta t/2, S + \Delta t \times k1/2)$$
$$k3 = f(t + \Delta t/2, S + \Delta t \times k2/2)$$
$$k4 = f(t + \Delta t, S + \Delta t \times k3)$$
$$S(next) = S + \Delta t/6 \times (k1 + 2k2 + 2k3 + k4)$$

Here f(t, S) = Qin(t) − Qout(H(S)), and the PondGate step is Δt = 2 s.

13.2 One step by hand

Take the base case at t = 1800 s (the time of the inflow peak). At this moment S = 65.608 m³, H = 0.3645 m, Qin = 0.1200 m³/s and Qout = 0.0748 m³/s.

Slope	Value (m³/s)
k1	0.045181
k2	0.045037
k3	0.045038
k4	0.044894

Substituting gives S_next = 65.608 + 2 ÷ 6 × (0.045181 + 2 × 0.045037 + 2 × 0.045038 + 0.044894) = 65.699 m³, and the water level rises to 0.3650 m. The four slopes are very close, which shows that a 2-second step is very small for this case.

13.3 Time-step sensitivity

Table 13-1 compares results for different steps from an independent program. RK4 hardly changes for steps up to 60 s. The simple forward Euler method loses accuracy

quickly as the step grows.

Method and step	Qp,o (m^3/s)	Hmax (m)
RK4, 2 s (PondGate)	0.09032	0.4132
RK4, 60 s	0.09028	0.4131
RK4, 300 s	0.08896	0.4091
Euler, 60 s	0.09179	0.4177
Euler, 300 s	0.09910	0.4396

The peaks for large steps in the table are also affected by sampling: with a 300-second step, the result is seen only every 5 minutes.

13.4 Mass-balance residual

The mass-balance residual ε/Vin checks whether "cumulative inflow − cumulative outflow" equals the change in storage. For the base case the absolute residual is 1.259×10^{-10} m^3, and the relative residual is shown as 0.000 %. Above 0.5 %, the results must not be used; check the inputs first.

13.5 How the numbers in this book are checked

Every example in Part II was run in the software by docs/book/run_examples.js, which read the results from the screen. docs/book/check_examples.py then recomputed them with an independently written RK4 program. The two agree on Qp,o, Hmax, ΔSmax and the drain-down residual in every case (Qp,o differs by less than 0.00005 m^3/s and Hmax by less than 0.0005 m). When one number is changed on purpose, the check program reports an error.

13.6 Exercises

1. Why are the outflow peaks for large steps too low? Hint: think about the sampling times.
2. If the mass-balance residual shows 2 %, which inputs should you check first?

Chapter 14 Example: how the weir width and pond area affect the results

14.1 Problem

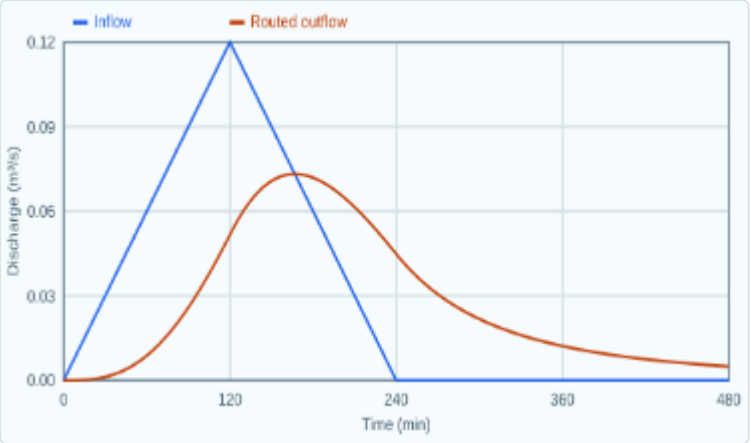
In the base case of Chapter 10 the outflow is too large (0.0903 m^3/s > 0.080 m^3/s). There are two ways to adjust it: change the weir width b, or change the pond area A. This chapter changes one variable at a time and watches how the results change.

14.2 Steps

1. Start from the base case (Start here → Book → press "Back to the base case", or reload the page).
2. In Module 3, change the weir width b to 0.10 m, press "Run routing and evaluation" and record the results.
3. Change b to 0.40 m, press "Run routing and evaluation" and record the results.
4. Press "Back to the base case" (b returns to 0.20 m). Then change the area A in Module 2 to 120 m^2 and press "Run routing and evaluation".
5. Change A to 360 m^2 and press "Run routing and evaluation" again.

14.3 Results

Change	Qp,o (m^3/s)	Hmax (m)	ΔSmax (m^3)	RQ
b = 0.10 m	0.0732	0.570	102.635	0.610
Base case b = 0.20 m, A = 180 m^2	0.0903	0.413	74.385	0.753
b = 0.40 m	0.1027	0.284	51.045	0.856
A = 120 m^2	0.1015	0.447	53.590	0.846
A = 360 m^2	0.0636	0.327	117.754	0.530



Hydrographs with a weir width of 0.10 m

Figure 14-1 shows the hydrographs for b = 0.10 m. Compared with Figure 10-1, the outflow is flatter and later.

14.4 Interpretation

- A narrower weir: the outflow falls, but the water level rises. The water is "held" in the pond, so more detention volume is needed.
- A larger area: both the outflow and the water level fall. The price is more land.
- Both b = 0.10 m and A = 360 m² bring Qp,o below 0.080 m³/s. With the first, the level rises to 0.570 m; with the second, only to 0.327 m. Which is better depends on the land cost and the allowable level.

This is the direction that the "Beginner check" box suggests: a larger pond area usually lowers both the peak water level and the peak outflow; a smaller outlet lowers the peak outflow but raises the peak water level.

14.5 Exercises

1. Using the base case, find the largest weir width that brings Qp,o just below 0.080 m³/s (to 0.01 m).
2. If Hallow = 0.50 m, which designs in the table above also meet Qallow = 0.080 m³/s?

Chapter 15 Example: rectangular openings and drain-down time

15.1 Problem

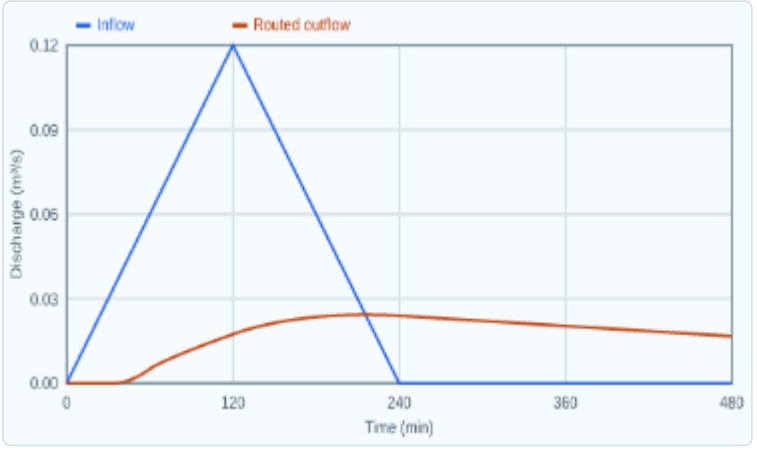
Replace the outlet of the base case with a rectangular opening (opening height 0.10 m, invert 0.05 m, Cd = 0.62) and compare three opening areas.

15.2 Steps

1. Start from the base case (press "Back to the base case"). In Module 3, untick "Sharp-crested weir" and tick "Rectangular opening".
2. Enter the opening area Ao as 0.005, 0.010 and 0.020 m² in turn, and press "Run routing and evaluation" after each one.

15.3 Results

Ao (m²)	Qp,o (m³/s)	Hmax (m)	ΔSmax (m³)	Storage left at Te + Td
0.005	0.0130	1.056	190.013	5.01 %
0.010	0.0243	0.936	168.476	5.41 %
0.020	0.0428	0.752	135.278	6.67 %



Hydrographs with an opening of 0.010 m²

Figure 15-1 shows the hydrographs for Ao = 0.010 m². The outflow is very flat and continues long after the event ends.

15.4 Interpretation

- All three openings keep the outflow peak far below 0.080 m^3/s, so the peak reduction is strong.
- But the water levels are all high. With Ao = 0.005 m^2 the level even exceeds the allowable 1.00 m.
- The drain-down matters even more: after 24 hours, more than 5 % of the event-added storage is still in the pond, above the declared 1 %. The main reason is the opening invert of 0.05 m. The water below the opening, 0.05 m × 180 m^2 = 9 m^3, can never flow out, and this is about 5 % of ΔSmax. In addition, the opening releases water very slowly at low levels (Chapter 12). The pond does not drain, so less volume is available when the next storm comes.

This chapter shows that looking only at the outflow peak can lead to a design that does not drain. This is why PondGate treats the drain-down residual as a screening limit. The "Beginner check" also lists the drain-down residual as its third item, and all three openings show "exceeds limit" there.

15.5 Exercises

1. When the opening area is doubled, by about what factor does Qp,o change? Why is it not exactly 2?
2. If the drain-down time Td is changed to 48 h, does the residual for Ao = 0.010 m^2 become larger or smaller? Check it in the software.

Chapter 16 Example: inflow shape and duration

16.1 Problem

Three inflows all have a peak of 0.120 m³/s. The pond and weir are those of the base case; only the inflow shape changes.

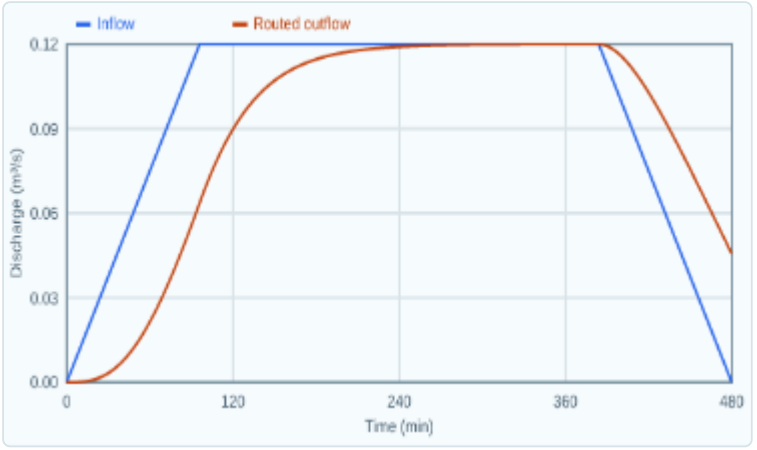
Inflow	Settings	Inflow volume
Short triangle (base case)	Tp = Tr = 30 min	216 m³
Long flat-topped	Rise 24, plateau 72, recession 24 min	691.2 m³
Long triangle	Tp = Tr = 96 min	691.2 m³

16.2 Steps

1. Long flat-topped: in Module 1, set "Standard case" to the long-duration event, keep the other defaults and press "Run routing and evaluation".
2. Long triangle: set the triangular event again, change both Tp and Tr to 96 min and press "Run routing and evaluation".

16.3 Results

Inflow	Qp,o (m³/s)	RQ	Hmax (m)	ΔSmax (m³)	FPI95
Short triangle	0.0903	0.753	0.413	74.385	0.006
Long flat-topped	0.1200	1.000	0.499	89.890	0.036
Long triangle	0.1123	0.936	0.478	86.018	0.011



Hydrographs of the long-duration event

Figure 16-1 shows the hydrographs of the long flat-topped event. The plateau lasts 72 minutes. The pond fills during the plateau, and the outflow catches up with the inflow.

16.4 Interpretation

- The peaks are the same, but the results differ a lot. The long flat-topped event has RQ = 1.000, which means this pond gives no peak reduction at all.
- The reason is the limited capacity of the pond. When the water keeps coming long enough, the pond fills to the equilibrium level where "outflow equals inflow", and the detention effect disappears.
- The long-duration events have a larger FPI95, which means the top of the outflow is flat. The "time of peak" then has little meaning; use the median-volume time and the discharge-centroid time instead.

So a design cannot rely on the peak discharge alone. For the same peak, check the most unfavourable duration and shape.

16.5 Exercises

1. In the long flat-topped event, what is the water level when the outflow equals the inflow? Work it back from the weir equation in Chapter 12.
2. If the plateau time is changed to 36 minutes, how does RQ change?

Chapter 17 Example: an irregular pond and a stage–storage table

17.1 Problem

Real ponds often have sloped sides or irregular shapes, so the area changes with the level. In that case, describe the pond with a stage–storage table. This example uses the data of Table 17-1; all other conditions are the base case.

Water level H (m)	0	0.25	0.50	0.75	1.00	1.50
Storage S (m^3)	0	30	80	150	240	420

This table is also in the download package as ch17_irregular_stage_storage.csv.

17.2 Steps

1. Start from the base case (press "Back to the base case"). In Module 2, set the storage representation to the tabulated stage–storage relation.
2. Paste Table 17-1 into the table field (one "level, storage" pair per line), or import the CSV file.
3. Press "Run routing and evaluation".

17.3 Results

Pond shape	Qp,o (m^3/s)	Hmax (m)	ΔSmax (m^3)
Constant area 180 m^2 (base case)	0.0903	0.413	74.385
Irregular pond (Table 17-1)	0.0912	0.416	63.161

17.4 Interpretation

The irregular pond is smaller at low levels (the mean area between 0 and 0.25 m is only 120 m^2). The water level rises faster, so the outlet starts to release a lot of water earlier. At 0.416 m the irregular pond holds only about 63 m^3, while the constant-area pond holds about 75 m^3. Less water stays in the pond and more is released, so Qp,o is slightly higher and the detention volume is about 11 m^3 smaller, while the peak levels are almost the same.

When you use a stage–storage table, note the following:

- The first row is usually the pond floor (S = 0).
- Routing does not extrapolate beyond the table. The highest level in the table must be above the highest possible water level; otherwise the result is invalid.
- The table and the outlet elevations must use the same datum.

17.5 Exercises

1. From Table 17-1, compute the mean area between 0.25 and 0.50 m.
2. What happens if the table only reaches H = 0.40 m?

Chapter 18 The full design workflow: from failing to feasible, and reading "Blocked"

18.1 Problem

In Chapter 14 you found a few feasible designs by manual trial and error. This chapter uses the constrained design screening of Module 5 instead, so the software searches the declared range automatically. The limits are the same as in Chapter 10: Qallow = 0.080 m^3/s, Hallow = 1.00 m and a residual of no more than 1 % within 24 hours.

18.2 Preparation: free-discharge evidence

The free-discharge evidence must be complete before constrained screening. This example is a teaching case, so the ticks and records are teaching assumptions.

1. Start from the base case (Start here → Book → press "Back to the base case").
2. In Module 3, tick "Free-discharge boundary confirmed" and the four checks.
3. In the "Free-discharge evidence record", enter: Teaching assumption (practice only): a single outlet with free discharge, not affected by tailwater, backwater, submergence or operational control.
4. In the "Screening-limit basis", enter: Teaching assumption (practice only): Qallow and Hallow are example values set in this book.

In a real project these two records must give the drawing or inspection number, the checker and the date, and the regulatory source of the limit values.

18.3 Example 1: minimum detention volume

1. In Module 5, enter Qallow = 0.080 and Hallow = 1.00, set the objective to "Minimum event-added detention volume ΔSmax", and keep the weir-width bounds at 0.05 to 1.00 m.
2. Press "Run constrained design screening".

Selected feasible constrained-screening case — routed

The program evaluated the full visible grid, then performed 2 local refinement passes inside the declared size bounds. The plots above are rerouted with this selected screening case. No screened outlet dimension remains at a declared bound.

Screening item	Result	Interpretation
Outlet configuration	Passive weir only — formal screening	Only listed outlet elements are active.
Plan area A	460 m²	Selected on the declared 20 m² grid.
Weir width b	0.53 m	screened
Event-added detention volume ΔSmax	92.0 m³	Maximum routed storage above the initial storage S0.
Drain-down residual at Te + Td	0.07 / 1.00%	Remaining event-added storage must satisfy the declared drawdown constraint.
Peak outflow / allowable	0.0798 / 0.0800 m³/s	Outlet-peak constraint satisfied.
Peak water level / allowable	0.200 / 1.000 m	Water-level constraint satisfied.
Response indicators	RQ = 0.665; RS = 0.426	Attenuation and event-added storage descriptors.

This is a conditional discrete screening result under the declared bounds. It is not a continuous global optimum or final engineering-design validation.

Screening result (minimum detention volume)

Figure 18-1 shows the screening result. The software searches areas from 60 to 600 m² (on a 20 m² grid) and the declared weir-width range, and then runs 2 passes of local refinement.

Item	Result
Plan area A	460 m²
Weir width b	0.53 m (the input field shows 0.525)
Event-added detention volume ΔSmax	92.0 m³
Peak outflow / allowable	0.0798 / 0.0800 m³/s
Peak water level / allowable	0.200 / 1.000 m
Drain-down residual	0.07 / 1.00 %

18.4 Example 2: minimum area

Change the objective to "Minimum plan area A" and press "Run constrained design screening" again.

Item	Result
Plan area A	100 m²
Weir width b	0.05 m (at the declared lower bound)
Event-added detention volume ΔSmax	94.6 m³
Peak outflow / allowable	0.0782 / 0.0800 m³/s
Peak water level / allowable	0.946 / 1.000 m

18.5 Reading the two results

- Both designs meet all limits, but the trade-offs are completely different. The minimum-volume design has very shallow water (0.200 m) and needs a large area. The minimum-area design needs only 100 m², but the water depth is close to the limit (0.946 m).
- In the minimum-area design the weir width stops at the lower bound of 0.05 m, and the screen marks it as a boundary case. This means the pond might be smaller still if a narrower weir were allowed. But a very narrow weir blocks easily, so the lower bound should follow the maintenance needs, not be relaxed to get a better number.
- Both results are only conditional screening within the declared range. Neither is a global optimum or a final design.

18.6 Example 3: Blocked

Make the limits very strict: Qallow = 0.040 m³/s and Hallow = 0.30 m. Set the objective back to minimum detention volume and press "Run constrained design screening" again.

The result says: "No feasible screening case was found within the declared physical bounds and discrete screening domain. Dominant constraints observed: peak-outflow limit, maximum-water-level limit, drain-down residual limit."

This block is reasonable. To bring a peak of 0.120 m³/s down to 0.040 m³/s, the pond must store a lot of water. But the level must not exceed 0.30 m, and even the upper area bound of 600 m² is not enough.

The right way to respond:

1. First check that the sources of Qallow and Hallow are correct. If they are, do not relax them just to get a solution.
2. Based on the dominant limits, consider adjustments with an engineering basis: for example, a larger area range if the land allows it, another outlet type, or several smaller facilities.
3. If a larger area is needed, use continuous optimization with a larger area range and check again.

18.7 Exercises

1. In Example 1, Qp,o = 0.0798 m^3/s, very close to 0.080. Why do screening results often lie close to a limit?
2. In Example 3, relax only Hallow to 1.00 m (Qallow stays at 0.040) and run it in the software. Is it still blocked?

Chapter 19 Cross-checks with HEC-HMS and SWMM

19.1 Why cross-check

Different programs use different numerical methods. If the inputs are exactly the same but the results differ a lot, at least one side has an error in its inputs, units or settings. A cross-check can find such errors, but it only shows "numerical consistency". It does not show that the results match field conditions.

19.2 Built-in example: E-W1 A-005

PondGate has a read-only verification case, E-W1 A-005. It is a laboratory-scale case with a single weir. The event lasts from 0 to 977.25 s, and the initial level is H0 = 0.07399 m. The stage–storage and outflow equations of the case are fixed on the page and do not change your normal routing inputs.

1. On the "Results, verification and records" page, find the "E-W1 A-005 external cross-check benchmark".
2. Press "Run E-W1 A-005 benchmark RK4".
3. Then press "Time-step sensitivity (0.125 / 0.250 / 0.500 s)".

19.3 Comparison of three engines

The three-engine table on the page compares the stored results of PondGate (DSS RK4, Δt = 0.25 s), EPA SWMM 5.2.4 and HEC-HMS 4.13 over a common window of 3,910 time points.

Engine	Peak outflow (m^3/s)	Peak time (s)
PondGate RK4	0.00613899	154.50
HEC-HMS 4.13	0.00613899	154.50
EPA SWMM 5.2.4	0.00614476	154.75

Pair	Outflow RMSE (m^3/s)
HEC-HMS and PondGate	8.57×10^{-9}
HEC-HMS and SWMM	2.30×10^{-6}
PondGate and SWMM	2.30×10^{-6}

The peaks of the three engines differ by less than 0.1 %. HEC-HMS uses the Modified Puls method. To match its one-minute computation interval, the case uses a volume-conserving conversion of time and discharge: one HEC minute stands for 0.25 s of the original record, the HEC discharge is 1/240 of the original discharge, and the output is converted back to the original scale.

19.4 Time-step sensitivity

RK4 step	Maximum outflow (m^3/s)	Mass residual (m^3)
0.125 s	6.13899×10^{-3}	2.49×10^{-10}
0.250 s	6.13899×10^{-3}	9.98×10^{-10}
0.500 s	6.13893×10^{-3}	-7.93×10^{-5}

When the step grows from 0.125 s to 0.500 s, the maximum outflow changes by only about 0.001 %, so a 0.25-second step is enough.

19.5 Compare your own case

1. Build the same pond in HEC-HMS or SWMM: the same inflow, initial level, stage–storage relation, outlet equation and coefficients.
2. Save the actual simulation results of the external model as a CSV. The first row holds the column names. The required columns are time_s (or time, t) and outflow_m3_s (or qout, qo); a full comparison also needs incremental_storage_m3. Press "Download external-output CSV template" for an example of the format.
3. In the "External-model cross check", choose the reference hydrograph (the current routing result or one of the public benchmarks A-005 to D-005) and the external model, import the CSV and press "Compare external result".

Common causes of disagreement are different units (L/s and m^3/s), different time bases (minutes and seconds), different datums and different coefficients in the external model.

19.6 Exercises

1. Why is the difference between HEC-HMS and PondGate much smaller than that between SWMM and PondGate? Hint: look at the data and methods each one

uses.

2. If the outflow peak of the external model is 240 times larger than that of PondGate, what is the most likely error?

Chapter 20 Capstone project: checking the detention pond of a site

This chapter brings the earlier chapters together in one complete check. The site and the numbers are a fictitious teaching case.

20.1 Case conditions

The post-development design inflow hydrograph of a development site is given in Table 20-1 (ch20_site_inflow.csv in the download package).

Time (min)	0	10	20	30	40	50	60	80	100	120
Discharge (m³/s)	0	0.03	0.09	0.21	0.25	0.18	0.12	0.06	0.025	0

Design limits (teaching assumption): Qallow = 0.15 m³/s, Hallow = 0.80 m, and a residual of no more than 1 % within 24 hours. The outlet is a sharp-crested weir with Cw = 1.70 and the crest at the pond floor.

20.2 Step 1: compute the inflow volume by hand

Use the trapezoidal rule: multiply the mean discharge of each pair of neighbouring points by the time interval, and add them up:

> $V_{in} = \Sigma\,(Q_i + Q_{i+1}) \div 2 \times \Delta t_i = 666\ m^3$

You can use this number to check whether the RS values later are reasonable.

20.3 Step 2: enter the hydrograph and try a design

1. In Module 1, set "Standard case" to the user-defined event and the inflow shape to imported time-discharge pairs. Paste Table 20-1 (one "time, discharge" pair per line).
2. In Module 2, enter A = 300 m²; in Module 3, enter the weir width b = 0.30 m.
3. In Module 5, enter Qallow = 0.15 and Hallow = 0.80.
4. Press "Run routing and evaluation" and read the "Beginner check".

Trial	A (m²)	b (m)	Qp,o (m³/s)	Hmax (m)	ΔSmax (m³)	Reading
1	300	0.30	0.2037	0.542	162.703	Outflow exceeds
2	500	0.30	0.1662	0.474	236.763	Outflow still exceeds

For trial 1, RS = 162.703 ÷ 666 = 0.244, the same as in the results table. A larger area lowers the outflow, but not enough, while the water level still has a large margin. From the lesson of Chapter 14, the weir should be made narrower at the same time.

20.4 Step 3: constrained screening

1. Complete the free-discharge evidence (teaching assumption, as in Chapter 18).
2. In Module 5, set the weir-width bounds to 0.10 to 0.60 m and the objective to "Minimum event-added detention volume ΔSmax".
3. Press "Run constrained design screening".

Selected feasible constrained-screening case — routed

The program evaluated the full visible grid, then performed 2 local refinement passes inside the declared size bounds. The plots above are rerouted with this selected screening case. No screened outlet dimension remains at a declared bound.

Screening item	Result	Interpretation
Outlet configuration	Passive weir only — formal screening	Only listed outlet elements are active.
Plan area A	520 m²	Selected on the declared 20 m² grid.
Weir width b	0.23 m	screened
Event-added detention volume ΔSmax	274.5 m³	Maximum routed storage above the initial storage S0.
Drain-down residual at Te + Td	0.17 / 1.00%	Remaining event-added storage must satisfy the declared drawdown constraint.
Peak outflow / allowable	0.1467 / 0.1500 m³/s	Outlet-peak constraint satisfied.
Peak water level / allowable	0.528 / 0.800 m	Water-level constraint satisfied.
Response indicators	RQ = 0.587; RS = 0.412	Attenuation and event-added storage descriptors.

This is a conditional discrete screening result under the declared bounds. It is not a continuous global optimum or final engineering-design validation.

Screening result of the capstone project

Figure 20-1 shows the screening result.

Item	Result
Plan area A	520 m²
Weir width b	0.23 m (the input field shows 0.225)
Event-added detention volume ΔSmax	274.5 m³
Peak outflow / allowable	0.1467 / 0.1500 m³/s
Peak water level / allowable	0.528 / 0.800 m
Drain-down residual	0.17 / 1.00 %

20.5 Step 4: check and record

1. Check that the mass-balance residual is close to 0 %.
2. Use the property from Chapter 11: the outflow peak should lie on the falling limb of the inflow.
3. Press "Download reproducible decision-support report" and file the report together with Table 20-1 and the basis of the limits.
4. Readers who are signed in can save this case as a project and share it with the colleague who reviews it.

20.6 Step 5: write the conclusion

A good conclusion gives the source of the inputs, the source of the limits, the selected sizes, the comparison with the limits and the scope of application. For example:

"For the design inflow of Table 20-1, Qallow = 0.15 m³/s and Hallow = 0.80 m, a detention pond with an area of 520 m² and a weir width of 0.23 m gives a peak outflow of 0.1467 m³/s, a peak water level of 0.528 m and a drain-down residual of 0.17 %, all within the limits. This result assumes free discharge and fixed coefficients. The freeboard, structural safety and blockage risk must be checked separately."

20.7 Exercises

1. If the site allows a pond of only 400 m², screen again with the minimum-area objective and see whether a feasible case exists.
2. Write a short conclusion that explains why the pond size cannot be decided from the peak inflow of 0.25 m³/s alone.

Appendix A Terms and symbols

Symbol	Term	Chinese term	Unit
Qp, Qp,i	Peak inflow	入流尖峰	m^3/s
Qp,o	Peak outflow	出流尖峰	m^3/s
Tp, Tr	Time to peak, recession time	洪峰到達時間、退水時間	min
T, Te	Event duration, event end	總事件延時、事件結束時間	min
Vin	Inflow volume	入流體積	m^3
S(H)	Stage–storage relation	水位—蓄水量關係	m^3
A	Plan area	平面面積	m^2
Hfloor	Floor elevation	池底高程	m
H0	Initial level	初始水位	m
Hmax	Peak water level	最高水位	m
ΔSmax	Event-added detention volume	事件新增滯洪量	m^3
Cw	Weir coefficient	堰流係數	—
b	Weir width	堰寬	m
P	Crest elevation	堰頂高程	m
Cd	Discharge coefficient	流量係數	—
Ao	Opening area	開口面積	m^2
RQ	Peak ratio Qp,o ÷ Qp,i	尖峰比	—
RS	Storage ratio ΔSmax ÷ Vin	蓄水比	—
RT	Normalized lag	無因次遲滯	—
FPI95	Near-peak flatness index	近洪峰平坦指標	—
ε/Vin	Mass-balance residual	質量平衡殘差	%
Qallow	Allowable peak outflow	容許出流尖峰	m^3/s
Hallow	Allowable water level	容許水位	m
Td	Maximum drain-down time	最長排空時間	h
Href	Reference level	參考水位	m
Δt	Time step	時間步長	s

For fuller explanations of the terms, see "Start here → Glossary" in the software.

Appendix B Error messages and what to do

Message or situation on screen	Meaning	What to do
The run button is grey, with a red list above it	A field is unreasonable	Click a field name in the list and fix the field outlined in red
"Screening not yet run"	Only normal routing was done	For normal routing read the "Beginner check"; to screen, go to Module 5
Module 5 lists missing conditions	The conditions for constrained screening are not met	Complete what the list says to check and to enter
"No feasible screening case was found ..."	No feasible design within the bounds	Adjust according to the dominant limits; see Chapter 18
Results are marked out of date	Inputs changed after the run	Press "Run routing and evaluation" again
Mass-balance residual above 0.5 %	A numerical problem	Check the units, the datum and the range of the stage–storage table
Importing time–discharge pairs fails	Format error	The first time is 0, times increase, discharges are not negative, at least two lines
No results after opening a project	A project stores only the inputs	Press "Run routing and evaluation" in Module 4
A viewer cannot save	Role setting	Ask the company administrator to change the role, or ask an engineer to save

Appendix C Answers to the exercises

All numerical answers were obtained by running this version of the software and were checked with an independent program. For explanation questions, the key points are given.

Exercise	Answer
10.1	Good enough (preliminary reading). Qp,o = 0.0903 ≤ 0.100 m³/s, Hmax = 0.413 ≤ 1.00 m and the drain-down residual 0.03 % ≤ 1 %. The beginner check says "Preliminary reading: all three limits are met".
10.2	A 0.20 m weir releases water too fast. The water flows away before it is stored, so the capacity of the pond is not fully used.
11.1	Vin = ½ × 0.120 × 5400 = 324 m³.
11.2	74.385 ÷ 180 = 0.413 m, equal to Hmax, because the floor and the initial level are both 0.
12.1	2 times. Weir flow is proportional to the weir width.
12.2	The opening rises from 0.0166 to 0.0253 m³/s, about 52 %. The weir rises from 0.1202 to 0.3400 m³/s, about 183 %.
13.1	A large step samples only at the end of each step, so it can miss the true peak. A larger step also gives a larger numerical error.
13.2	The units (L/s and m³/s, minutes and seconds), the datum, whether the stage–storage table covers the highest level, and any extreme or unreasonable input.
14.1	With b = 0.12 m, Qp,o = 0.0780 m³/s (Hmax = 0.527 m). With b = 0.13 m, Qp,o = 0.0801 m³/s, which exceeds the limit. So the largest weir width is 0.12 m.
14.2	Only A = 360 m² (Qp,o = 0.0636 m³/s, Hmax = 0.327 m). For b = 0.10 m, Hmax = 0.570 m, which exceeds 0.50 m.
15.1	It changes from 0.0243 to 0.0428 m³/s, about 1.76 times. With a larger opening the water level in the pond is lower, so the head is smaller, and the discharge of an opening grows only with about the 0.5 power of the head.
15.2	It becomes only slightly smaller: from 5.41 % to 5.36 %. The 9 m³ below the opening invert can never drain, so a longer drain-down time still cannot meet the 1 % limit.
16.1	0.120 = 0.34 × H^1.5, so H = (0.120 ÷ 0.34)^(2/3) = 0.499 m, exactly the Hmax of the software.
16.2	Qp,o = 0.1191 m³/s, RQ = 0.993, Hmax = 0.497 m. Even with half the plateau, the peak reduction is still less than 1 %, because 36 minutes is still long enough for the pond to approach the equilibrium level.
17.1	(80 − 30) ÷ 0.25 = 200 m².
17.2	If the peak level goes beyond the table, routing does not extrapolate, the S(H) boundary check shows a block and the result must not be used. Add data for higher levels.

Exercise	Answer
18.1	The objective is the minimum detention volume. A larger outflow means less water must be stored, so the best point pushes the outflow up to just below Qallow.
18.2	It is not blocked. Screening finds A = 260 m² and b = 0.05 m (at the lower bound), with Qp,o = 0.0393 m³/s, Hmax = 0.598 m and a drain-down residual of 0.70 %. The block in Example 3 came mainly from Hallow = 0.30 m; but limits may only be changed on a regulatory or design basis.
19.1	HEC-HMS uses the same stage–storage and outflow tables as PondGate, so the difference comes only from the numerical method. SWMM uses its own storage curve and outlet representation, so the difference is larger, but the peaks still differ by less than 0.1 %.
19.2	Most likely the time and discharge scale conversion (laboratory scale factor 240) was not applied. Check the comparison-scale setting and the units of the external model.
20.1	Yes. The minimum-area objective selects A = 380 m² and b = 0.12 m (the input field shows 0.123), with Qp,o = 0.1392 m³/s, Hmax = 0.761 m and a drain-down residual of 0.20 %. With a smaller area, the water level is closer to the 0.80 m limit.
20.2	Key point: for the same peak, a different inflow duration and shape give a different inflow volume (Chapter 16). In this case the inflow volume is 666 m³, and the storage needed depends on both the volume and the release capacity of the outlet.

Appendix D References and software citation

- PondGate source code: GitHub hongyaoming-cloud/detention-pond-dss (MIT licence).
- PondGate v1.0.0: DOI 10.5281/zenodo.23296722.
- PondGate v1.0.1: DOI 10.5281/zenodo.23298538.
- PondGate v1.0.2: DOI 10.5281/zenodo.23299011.
- PondGate, all versions (concept DOI): 10.5281/zenodo.23296721, which resolves to the latest version.
- U.S. Army Corps of Engineers, HEC-HMS 4.13 user's manual (Modified Puls routing).
- U.S. EPA, Storm Water Management Model (SWMM) 5.2 user's manual.

Suggested citation: Hong, Y.-M. PondGate: open-source detention-pond routing software (version 1.3.4). Zenodo. Use the version DOI shown on the Zenodo page. Related journal papers will be added to a later edition after publication.